

Problem Set #1

EH6127 - Quantitative Methods

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This homework makes use of data available in `{stevedata}` and implies the use of `{tidyverse}` to answer the questions. `{tidyverse}` is not necessary to answer these questions though it will assuredly make the process easier. Load these two libraries to get started answering these questions.

```
library(tidyverse)
library(stevedata)
```

Systemic Banking Crises Database II

This homework assignment will refer to the SBCD data set that is available in `{stevedata}`. This data set contains a (very) basic summary of banking, currency, debt, and debt-restructuring crises from 1970 to 2017 and are made available by Luc Laeven and Fabian Valencia (2020) in the *IMF Economic Review*. You can find out more information about the data by visiting [this part of the package's website](#), or with the following command.

```
?SBCD
```

Here's a little preview of these data.

```
SBCD
```

```
## # A tibble: 547 x 4
##   country type          year month
##   <chr>   <chr>        <dbl> <int>
## 1 Albania banking      1994    NA
## 2 Albania currency     1997     1
## 3 Albania debt         1990    NA
## 4 Albania debtrestructuring 1992    NA
## 5 Algeria banking      1990    NA
## 6 Algeria currency     1988    11
## 7 Algeria currency     1994     4
## 8 Angola  currency     1991     3
## 9 Angola  debt         1988    NA
## 10 Angola debtrestructuring 1992    NA
## # i 537 more rows
```

Answer these questions. A successful answer of these question must include the R code you used to help you answer the question. Each question is worth a point.

1. Which country appears the most in these data?
2. How many distinct countries are there in these data?
3. What is the most common type of crisis to occur in these data? What is the least common?
4. Which year had the most crises?
5. How many crises were there in 2008, the height of the Great Recession?
6. There are three occurrences for Sweden in these data. Reproduce these observations for me.
7. Subset the data to just the crises involving Denmark, Finland, Iceland, Norway, and Sweden. Assign this subset of the data to a new object called `Nordics` and reproduce these observations here.
8. The `month` variable is mostly missing data as the authors do not often record the month of the crisis. If I wanted to get rid of this column, how might I do that?

Extra Credit (1 Point Each)

9. Suppose I wanted to create a Cold War dummy variable that equals 1 if the crisis occurred on or before 1990 and equals a 0 if it occurred afterward. How might I do that?
10. These data are structured 1) alphabetically by English country name, then 2) by type of crisis in alphabetical order and, finally, 3) by year. For each country in the data, show me how you could create a data/table that lists the first occurrence of a banking (i.e. `type = 'banking'`) crisis.